

An Introduction to the Use of Finite Element Procedures

1.1 INTRODUCTION

Finite element procedures are at present very widely used in engineering analysis, and we can expect this use to increase significantly in the years to come. The procedures are employed extensively in the analysis of solids and structures and of heat transfer and fluids, and indeed, finite element methods are useful in virtually every field of engineering analysis.

The development of finite element methods for the solution of practical engineering problems began with the advent of the digital computer. That is, the essence of a finite element solution of an engineering problem is that a set of governing algebraic equations is established and solved, and it was only through the use of the digital computer that this process could be rendered effective and given general applicability. These two properties—effectiveness and general applicability in engineering analysis—are inherent in the theory used and have been developed to a high degree for practical computations, so that finite element methods have found wide appeal in engineering practice.

As is often the case with original developments, it is rather difficult to quote an exact “date of invention,” but the roots of the finite element method can be traced back to three separate research groups: applied mathematicians—see R. Courant [A]; physicists—see J. L. Synge [A]; and engineers—see J. H. Argyris and S. Kelsey [A]. Although in principle published already, the finite element method obtained its real impetus from the developments of engineers. The original contributions appeared in the papers by J. H. Argyris and S. Kelsey [A]; M. J. Turner, R. W. Clough, H. C. Martin, and L. J. Topp [A]; and R. W. Clough [A]. The name “finite element” was coined in the paper by R. W. Clough [A]. Important early contributions were those of J. H. Argyris [A] and O. C. Zienkiewicz and Y. K. Cheung [A]. Since the early 1960s, a large amount of research has been devoted to the technique, and a very large number of publications on the finite element method is

available (see, for example, the compilation of references by A. K. Noor [A] and the *Finite Element Handbook* edited by H. Kardestuncer and D. H. Norrie [A]).

The finite element method in engineering was initially developed on a physical basis for the analysis of problems in structural mechanics. However, it was soon recognized that the technique could be applied equally well to the solution of many other classes of problems. The objective of this book is to present finite element procedures comprehensively and in a broad context for solids and structures, field problems (specifically heat transfer), and fluid flows.

To introduce the topics of this book we consider three important items in the following sections of this chapter. First, we discuss the important point that in any analysis we always select a *mathematical model* of a physical problem, and then we solve *that* model. The finite element method is employed to solve very complex mathematical models, but it is important to realize that the finite element solution can never give more information than that contained in the mathematical model.

Then we discuss the importance of finite element analysis in the complete process of computer-aided design (CAD). This is where finite element analysis procedures have their greatest utility and where an engineer is most likely to encounter the use of finite element methods.

In the last section of this chapter we address the question of how to study finite element methods. Since a voluminous amount of information has been published on these techniques, it can be rather difficult for an engineer to identify and concentrate on the most important principles and procedures. Our aim in this section is to give the reader some guidance in studying finite element analysis procedures and of course also in studying the various topics discussed in this book.

1.2 PHYSICAL PROBLEMS, MATHEMATICAL MODELS, AND THE FINITE ELEMENT SOLUTION

The finite element method is used to solve physical problems in engineering analysis and design. Figure 1.1 summarizes the process of finite element analysis. The physical problem typically involves an actual structure or structural component subjected to certain loads. The idealization of the physical problem to a mathematical model requires certain assumptions that together lead to differential equations governing the mathematical model (see Chapter 3). *The finite element analysis solves this mathematical model.* Since the finite element solution technique is a numerical procedure, it is necessary to assess the solution accuracy. If the accuracy criteria are not met, the numerical (i.e., finite element) solution has to be repeated with refined solution parameters (such as finer meshes) until a sufficient accuracy is reached.

It is clear that the finite element solution will solve only the selected mathematical model and that all assumptions in this model will be reflected in the predicted response. We cannot expect any more information in the prediction of physical phenomena than the information contained in the mathematical model. Hence the choice of an appropriate mathematical model is crucial and completely determines the insight into the actual physical problem that we can obtain by the analysis.

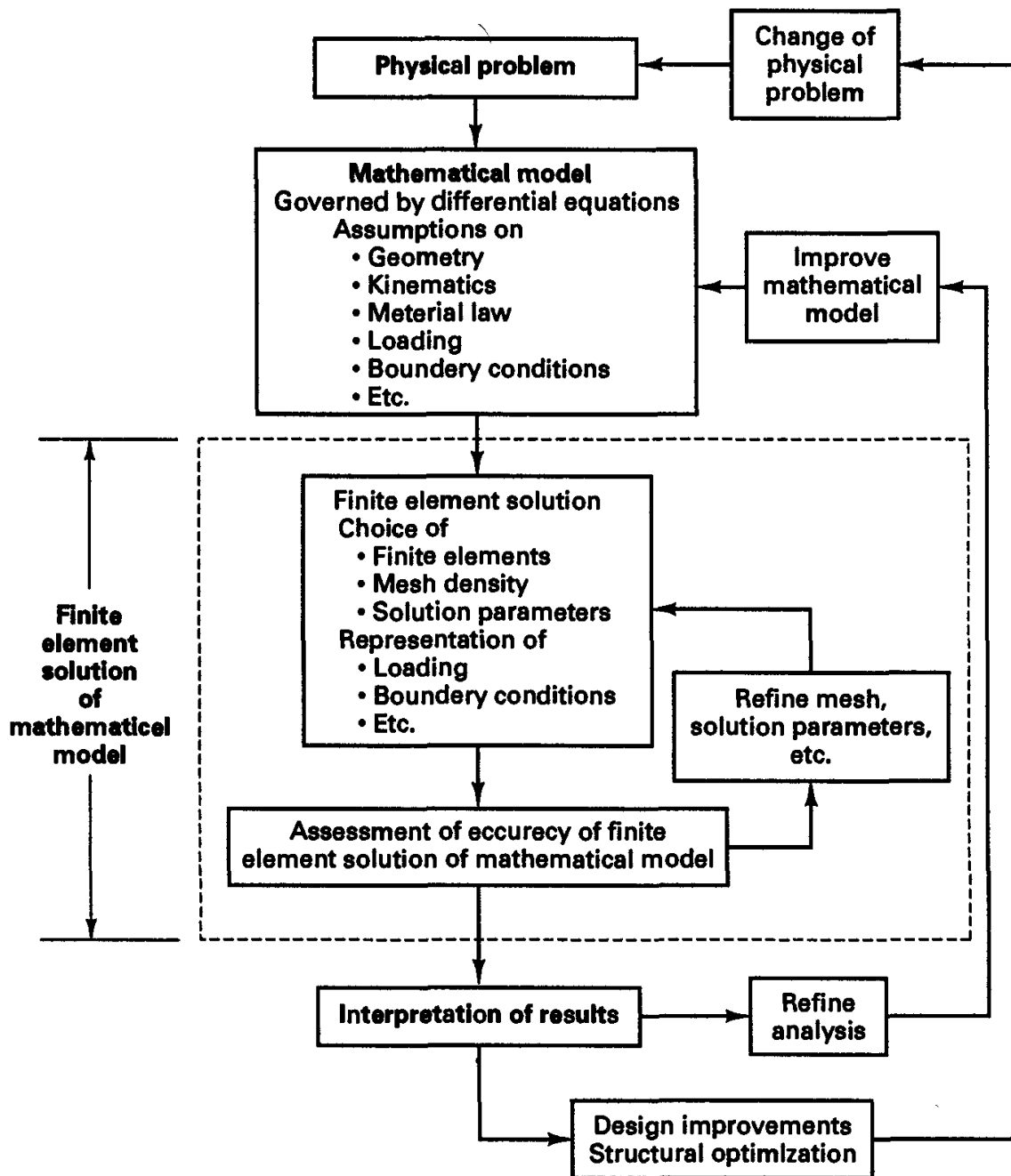


Figure 1.1 The process of finite element analysis

Let us emphasize that, by our analysis, we can of course only obtain *insight* into the physical problem considered: we cannot predict the response of the physical problem *exactly* because it is impossible to reproduce even in the most refined mathematical model all the information that is present in nature and therefore contained in the physical problem.

Once a mathematical model has been solved accurately and the results have been interpreted, we may well decide to consider next a refined mathematical model in order to increase our insight into the response of the physical problem. Furthermore, a change in the physical problem may be necessary, and this in turn will also lead to additional mathematical models and finite element solutions (see Fig. 1.1).

The key step in engineering analysis is therefore choosing appropriate mathematical models. These models will clearly be selected depending on what phenomena are to be

predicted, and it is most important to select mathematical models that are *reliable* and *effective* in predicting the quantities sought.

To define the reliability and effectiveness of a chosen model we think of a *very-comprehensive* mathematical model of the physical problem and measure the response of our chosen model against the response of the comprehensive model. In general, the very-comprehensive mathematical model is a fully three-dimensional description that also includes nonlinear effects.

Effectiveness of a mathematical model

The most effective mathematical model for the analysis is surely that one which yields the required response to a sufficient accuracy and at least cost.

Reliability of a mathematical model

The chosen mathematical model is reliable if the required response is known to be predicted within a selected level of accuracy measured on the response of the very-comprehensive mathematical model.

Hence to assess the results obtained by the solution of a chosen mathematical model, it may be necessary to also solve higher-order mathematical models, and we may well think of (but of course not necessarily solve) a sequence of mathematical models that include increasingly more complex effects. For example, a beam structure (using engineering terminology) may first be analyzed using Bernoulli beam theory, then Timoshenko beam theory, then two-dimensional plane stress theory, and finally using a fully three-dimensional continuum model, and in each case nonlinear effects may be included. Such a sequence of models is referred to as a hierarchy of models (see K. J. Bathe, N. S. Lee, and M. L. Bucelem [A]). Clearly, with these hierarchical models the analysis will include ever more complex response effects but will also lead to increasingly more costly solutions. As is well known, a fully three-dimensional analysis is about an order of magnitude more expensive (in computer resources and engineering time used) than a two-dimensional solution.

Let us consider a simple example to illustrate these ideas.

Figure 1.2(a) shows a bracket used to support a vertical load. For the analysis, we need to choose a mathematical model. This choice must clearly depend on what phenomena are to be predicted and on the geometry, material properties, loading, and support conditions of the bracket.

We have indicated in Fig. 1.2(a) that the bracket is fastened to a very thick steel column. The description “very thick” is of course relative to the thickness t and the height h of the bracket. We translate this statement into the assumption that the bracket is fastened to a (practically) rigid column. Hence we can focus our attention on the bracket by applying a “rigid column boundary condition” to it. (Of course, at a later time, an analysis of the column may be required, and then the loads carried by the two bolts, as a consequence of the load W , will need to be applied to the column.)

We also assume that the load W is applied very slowly. The condition of time “very slowly” is relative to the largest natural period of the bracket; that is, the time span over which the load W is increased from zero to its full value is much longer than the fundamental period of the bracket. We translate this statement into requiring a static analysis (and not a dynamic analysis).